VX-One: Typed, Calibrated Decision Heads for Real-Time Perimeter Triage

An empirical study of unknown-attack honesty, cascade gray zones, and calibration under drift

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Abstract

Perimeter triage faces a latency/semantics trade-off: signature engines are fast and format-safe but blind to unseen attacks, while generative models are semantic but slow and produce unparseable output. In late 2026 a class of commercial *System One* decision models (TypeSafe Jev; Microsoft-Decision-1) began selling typed decision primitives—binary gates, closed-set choices, ordinal scores—as product APIs, with no academic treatment or independent evaluation. We present *VX-One*: a small multi-head network (100K–10M parameters) whose outputs *are* these primitives—a calibrated *gate*, a *choice* over a closed candidate set that always includes an explicit unknown option, and an ordinal *score* (conditional ordinal regression), trained multi-task, temperature-calibrated per head (~0.7–1.7 ms single-event latency p50 on CPU). We evaluate it on three public benchmarks (UNSW-NB15, CIC-IDS2017, EMBER2024) against calibrated XGBoost, Random-Forest, MLP, and Kitsune baselines, and measure a two-phase cascade (Suricata with 52,674 Emerging-Threats rules → VX-One) over five days of CIC-IDS2017. Our findings are deliberately double-sided: (i) VX-One wins *detection* (AUROC/FPR@95%TPR) against every calibrated baseline on UNSW-NB15 and under temporal drift on CIC-IDS2017 (0.711 vs. 0.637), but calibrated XGBoost wins multi-class *attribution* on all three benchmarks; (ii) under leave-one-family-out the gate detects never-seen families on 9/9 UNSW folds yet drops below chance on CIC's *recon* family, and an abstention threshold calibrated on one benchmark transfers poorly to another; (iii) on payload-less reconstructions of the CIC-IDS2017 days, signature engines resolve only ~0.1% of daily flows—the *gray zone* under payload opacity is ~99.9% of the traffic, which Phase 2 recovers (combined recall ≥0.978); (iv) probability calibration collapses under temporal drift (ECE 0.04→0.40) for *every* method while ranking partially survives. We contribute not novel estimators but a measured account of where typed calibrated decisions help perimeter triage, and where they fail.

Keywords

intrusion detection, calibrated decisions, out-of-distribution detection, malware classification, two-phase triage

1 Introduction

Operating a mitigation policy on the network boundary requires, simultaneously: nanosecond-scale decisions on the line of fire, calibrated confidences that a controller can threshold, and honest behavior on events that training has never seen. Three incumbent technology classes each miss at least one requirement.

(1) **Deterministic engines** (Snort [31], Suricata [27], Zeek/Bro [29], YARA [40]) are fast and format-safe but structurally blind to zero-day and low-and-slow offensives; the recurring failure mode is a rule explosion that no operations team can maintain.
(2) **Text LLMs for security** (e.g., SecureBERT [1])— *bidirectional encoders, not autoregressive models*, a conflation we correct—bring domain semantics, but as fine-grained decision engines on the line of fire they are heavy, and decoder-style LLM agents add generation latency and output-format fragility that a mitigation controller cannot tolerate.
(3) **Classical ML for intrusion detection** performs well on tabular attribution—XGBoost [6] wins our attribution benchmarks—but treats every event as a membership test, producing confident misattributions on families the model never saw (Section 5.2).

During October 2026, a commercial class of models positioned exactly at this gap began to ship: *System One* decision models, sold as APIs that take a structured `state` plus typed questions and return calibrated option probabilities (TypeSafe's Jev [36, 37]; Microsoft-Decision-1 [22, 23]). No peer-reviewed academic literature covers the class, its evaluation is vendor-reported, and no security-domain assessment exists.

This paper occupies the gap *by building the primitives ourselves* rather than remoting into a vendor API. VX-One maps a fixed-size event state to three typed decisions in a single forward pass, then evaluates the whole pipeline—including a real signature engine in front of it—on public data with anti-strawman baselines. The contributions are:

- **A typed decision engine** (Section 3): shared encoder + three primitive heads—*gate* (calibrated sigmoid), *choice* (softmax over a declared closed set that *always contains* unknown), *score* (CORN conditional ordinal regression [34])—with per-head temperature calibration on validation data, trained multi-task with a homoscedastic weighting scheme [17].
- **Unknown-attack honesty, measured** (Section 5.2): leave-one-attack-family-out across 9 UNSW-NB15 and 7 CIC-IDS2017 families, plus 4 months of EMBER2024 family drift, reporting both unseen-family detection (gate AUROC) and *attribution honesty* (how often the unknown option absorbs what the model should not claim to know).
- **A measured gray zone** (Section 5.4): Suricata 7 with 52,674 live Emerging-Threats rules over flow-reconstructed CIC-IDS2017 traffic resolves only ~0.1% of daily flows under payload opacity; the combined cascade recovers it with

recall ≥ 0.978, at the cost of a measured precision drop on web-attack days where Phase 1 is entirely blind.
- **Negative results, measured** (Sections 5.1–5.5): calibration collapses under drift for every method (ECE 0.04⟶0.40); class-weight tilts that help weak-label attribution can *hurt* it on saturated benchmarks; a fixed abstention threshold is not portable across calibrations; dropping the `unknown` option raises macro-F1 by up to +11.2 points—the honesty/utility trade-off is real and quantifiable.

Our heads are standard estimators; we say so. The claim is about their *packaging and evaluation*: event-level honesty instead of confident misattribution, measured end-to-end on data with audited labels [11, 32].

2 Related Work

System One decision models. Microsoft's announcement [22] and documentation [23] define the primitive triad (binary gate / closed-set choice / ordinal score) with calibrated option probabilities and single-pass scoring; TypeSafe's Jev [36, 37] commercialized the framing earlier in 2026. Both are vendor announcements without peer-reviewed evaluation; benchmarks cited (193.6× faster, 444.6× cheaper) are industry claims. Our work is the first academic instantiation and security-domain evaluation of the triad; we use their *names* only for attribution, define our own formal objects (Section 3), and deliberately avoid inheriting vendor terminology as if it were neutral vocabulary.

Single-pass inference. "Non-autoregressive" originates in neural machine translation [14]: full output in one forward pass, no decoding loop. We use the term strictly for architecture (shared trunk, parallel typed heads). We do not claim novelty from single-pass-ness—BERT-family classifiers are also single-pass—and our baselines include appropriate encoders; the differentiator is typed calibrated outputs, not one-pass-ness alone. Fast early-exit decision layers for per-event actions predate this design in spirit—the Viola–Jones cascade [39] is their ancestor.

Calibration and abstention. Platt scaling [30], multiclass probability transforms [42], empirical per-family calibration gaps [26], and temperature scaling [15] define our post-hoc protocol; conformal prediction [3, 41] is an optional wrapper we defer, noting that distribution-free coverage claims need exchangeability that temporal splits violate [4]. Chow's reject option [7] and selective classification [12] ground our abstention semantics: with symmetric rejection cost the Bayes-optimal threshold is $1 - \lambda_r = 0.5$, which is the threshold we initially set, and we measure how poorly it transfers where the model is overconfident (Section 5.2).

Signature engines and their limits. Snort [31], Zeek [29], YARA [40], and Suricata [27] anchor the deterministic tier. The DARPA-critique tradition [20, 21] set the standard for dataset skepticism; audits of CICIDS2017 [11, 32] document systematic labeling errors. We use both stratified and temporal splits and audit against the corrected labels.

ML on network traffic and malware. Kitsune [24] is the canonical packet-level unsupervised ensemble; we adapt it to flows (documented) and report its Platt-calibrated but weakly discriminative AUROC (≈0.61) as a concrete instance of "calibration creates no signal". Boosted trees [6] are the attribution baseline our engine *loses to* on all three benchmarks—we calibrate them with the same protocol to make the comparison non-strawman. EMBER2018 [2] and EMBER2024 [5], UNSW-NB15 [25], and CIC-IDS2017 [33] supply the data tiers. Viola–Jones [39] is the ancestor of our second-phase-only-when-needed cascade.

3 VX-One Design

3.1 State object

Each telemetry event is compiled to a flat state

$$S = \left(x_{\text{num}} \in \mathbb{R}^d,\ x_{\text{txt}}\right),$$

where $d \in \{39\ (\text{UNSW}), 77\ (\text{CIC-IDS2017}), 2568\ (\text{EMBER2024})\}$ and x_{txt} are protocol-level categorical fields (proto, service, state) when the benchmark provides them (UNSW only). The feature stage ϕ (median imputation of non-finite values with a row flag; signed log $1p$; robust scaling by train median/IQR; clip to ±8) normalizes arbitrarily heavy-tailed benchmark features into a fixed-size vector. Everything is byte-level reproducible from frozen JSON configs with a single `SEED=42`.

3.2 Trunk and typed heads

A small pre-norm MLP trunk ($L \in \{2..8\}$ blocks of [Linear → LayerNorm → GELU → Dropout 0.1], residual from block 2) maps $\phi(S)$ plus text-lookup embeddings to a shared representation $h = f_\theta(S)$, feeding four heads—one per typed primitive:

$$\begin{aligned}
&\textit{gate}: && \text{gate}(S) = \sigma(w_1^\top h + b) \in [0,1]; && \text{(Gate)}\\
&\textit{choice}: && \mathbf{p} = \text{softmax}(W_C h), \quad C \ni \text{unknown}; && \text{(Choice)}\\
&\textit{score}: && \pi_k = \prod_{i \le k} \sigma_q(h), && \text{(Score)}
\end{aligned}$$

where the Score head is conditional ordinal regression (CORN [34]) over four severity levels—benign, low, high, critical—and the choice head is a closed candidate set declared per dataset. We deliberately use *neutral* head names rather than vendor naming conventions. Trunk depth/width by scale (L, w): 2/224 (Nano), 3/576 (Small), 4/1280 (Base), with reduced input projections on EMBER2024 (whose 2568-dim feature projection dominates the parameter budget); verified parameter counts on UNSW-NB15 are 97K / 914K / 5.9M, inside a hard 100K/1M/5–10M budget that is enforced by unit test. The choice head's open-set semantics are *hybrid*: where the training data offers a real `unknown` population (EMBER2024's 2.19M-file unknown ground truth), the option is a trained class; where it does not (UNSW, CIC), unknown is realized as abstention at decision time via $(\max_k p_k < \tau_{\text{abs}})$, with τ_{abs} and any composed policy belonging to the mitigation controller, not the network.

3.3 Training formulation

Multi-task loss

$$\mathcal{L} = \sum_q \mathcal{L}_q,$$

with a modified homoscedastic balancing [17]: per-task reference uncertainty with a modified per-task reference uncertainty so that task-scale differences ($\ln K$ vs. $\ln 2$ for Choice vs. Gate) do not

dominate the initialization. Per-class loss weights on Choice use sqrt-normalized, mean-1 tilts ω_k with an *analytic de-tilt* in inference ($z \mapsto z - \log \omega$) applied before calibration, so the recovered posterior remains population-correct. The gate loss is *unweighted* BCE: weighting shifts the optimal logit by $\log(\omega_+/\omega_-)$, an offset temperature calibration cannot undo. Training uses AdamW (weight decay 10^{-2} on 2D weight matrices only), linear warmup (5%) + cosine decay, gradient clipping at 1.0, mixed precision, and early stopping (patience 4) on validation loss. All backbone hyperparameters are frozen across datasets and scales (D11-like discipline).

3.4 Decision controller semantics

The controller composes three binary gates from calibrated head outputs: the trained gate ($p^{\text{cal}} \geq \tau_{\text{det}}$, default 0.5); a drop gate on the exact CORN marginal $C_2 = \mathbb{P}(\text{sev} \geq 2 \mid x)$; and a low-urgency gate on the `recon` coordinate of the choice head. The severity rubric makes both implications exact—sev $\geq 2 \implies$ attack and recon $\implies$ sev=1—and they are verified with zero rule violations on 3.09M labeled flows, so each gate's target event coincides with exactly the event its own head estimates.

Gates are single-head functionals, by design. For single-head functionals, post-processing (de-tilt, temperature, thresholding) keeps the calibration error within a bound determined by the head's own deficiency (Appendix A, Lemma 1). Across heads, *no useful composition bound exists*: two perfectly-calibrated heads can combine into a joint policy whose calibration error is 1/4 in the worst case, whatever the marginal ECEs say (Appendix A, Prop. 4)—and the *A=B* case that attains it (two heads estimating the same underlying question) is the realistic one in security. The controller therefore composes *boolean decisions* between gates and never *probabilities* between heads; any future composed gate must be rubric-reducible to a single head, or measure its own joint calibration.

Thresholds are cost statements, not constants. With uniform misclassification cost and escalation cost λ_r, the Bayes rule with rejection abstains iff $\max_k p_k < 1 - \lambda_r$ [7, 8]: $\tau_{\text{abs}} = 0.5$ *is* the statement $\lambda_r = 0.5$ (unknown costs as much as a wrong attribution), and $\tau_{\text{det}} = 0.5$ *is* the statement that missing an attack costs the same as a false alarm. For operators who hold different cost ratios, the mapping is the standard cost-sensitive one, $\tau^* = c_{\text{FP}}/(c_{\text{FP}}+c_{\text{FN}})$ [10]—e.g., a 10:1 miss:alarm ratio implies $\tau^* \approx 0.091$; we keep 0.5 as the declared default and expose the mapping rather than tune silently. General policies with a full cost matrix live in the controller—max-prob abstraction can deviate from true Bayes when costs are asymmetric (an event with $\max_k p_k$ above threshold may have lower expected cost elsewhere)—and we measure the deviation of the abstraction on validation data rather than assume it away.

4 Experimental Setup

Data. UNSW-NB15 [25]: 149,039 / 26,302 / 82,332 flows (official split); 9 attack families. *CIC-IDS2017* [33] (MLCVE-corrected variant): 2.83M flows; two splits, stratified (sanity) and by-day (temporal drift). *EMBER2024* [5]: 3.85M binaries; temporal split by week, with the public challenge set held out; 2.19M ground-truth-unknown training rows. Label errors in the original CICIDS2017 are documented [11, 32]; on top of the audited variant, we repair raw-label encoding artifacts (U+FFFD mojibake in the web-attack labels) and a singular/plural naming drift (`Backdoor`), which a naive ingestion would silently route to the `unknown` class; micro-classes (11 Heartbleed, 36 Infiltration flows) are retained rather than dropped and stratified exactly. Every count is checked against the frozen schema in code before any experiment runs.

Baselines. XGBoost (500 boosted trees, hist mode), Random Forest (300 trees, min-leaf 1), an MLP (2×256), and Kitsune (9-autoencoder ensemble with feature bagging, Platt-calibrated on validation)—all trained with the *same* feature stage, the same per-class loss weights where analogues exist, and the *same* per-head temperature-scaling calibration as VX-One. This is anti-strawman discipline: boosted trees *benefit* from the calibration fix [26], and we report them raw *and* calibrated. A zero-shot API baseline from the commercial System One class (Section 5.6) was planned; it was not executed this cycle.

Metrics. Detection: AUROC and **FPR@95%TPR** (the FPR at the threshold capturing 95% of true attacks—an operational line-of-fire number that accuracy-style metrics hide). Attribution: label-aware macro-F1 (GT-unknown rows excluded by label, as they have no known counterpart). Calibration: 15-bin ECE, MCE, Brier, NLL, and reliability diagrams, reported on raw and calibrated heads. Latency: single-thread, batch=1, 3,000 timed events, model-exact hardware recorded with each result.

Infrastructure. Training on serverless GPUs (L4) with atomic checkpointing and tested resumption; evaluation computed *locally* from dumped raw logits so that all metric code is inspectable. 60 unit tests including end-to-end smoke training (`pytest`, 60/60 green).

5 Results

5.1 E1/E3 — Detection, attribution, calibration

Table 1 summarizes the two-tier comparison on test (calibrated temperatures). Three findings:

(i) VX-One wins detection, not attribution. On UNSW-NB15 and temporal CIC, VX-One tops AUROC and FPR@95%TPR. On CIC by-day *every* method's achievable FPR@95 is degraded, but the trees' degradation is total (FPR@95 = 1.0 — no threshold reaches 95% TPR at nontrivial FPR), while VX-One retains a partially usable operating point. Attribution (macro-F1) inverts this: calibrated XGBoost wins on UNSW, stratified CIC, and EMBER2024—by 4–8 points. *We report the inversion rather than hide it*: our contribution does not rest on attribution wins, and we are explicit about the loss.

(ii) Saturation makes stratified CIC a sanity benchmark, not a selling point. All methods land at ∼1.0 with ECE ≤0.001. Headline numbers on saturated benchmark splits are best treated as pipeline-correctness checks.

(iii) Kitsune: calibration does not equal signal. With Platt calibration on validation, Kitsune is well-calibrated (ECE 0.037) yet near-chance on discrimination (AUROC 0.608). Its ensemble-of-autoencoders signal does not survive hostile-stage re-fit to benign-only data—we re-fit *its* feature stage to its own training distribution (benign only), as a real unsupervised deployment would. Platt on a useless score calibrates a useless probability.

Table 1: Test-set comparison (calibrated). VX: VX-One at three sizes. FPR@95 is FPR@95%TPR. † = under temporal drift. [a] = attribution benefiting from drift; [b] = detection plateau of saturation (all models ≈ 1.0). Full per-dataset table in the released results.

Dataset	Model	AUROC	FPR@95	macro-F1
UNSW-NB15	VX Small	**0.9834**	**0.1211**	0.4571
	XGBoost	0.9818	0.1369	**0.4968**
	RF	0.9792	0.1399	0.4378
	MLP	0.9821	0.1279	0.4445
	Kitsune	0.6086	0.5617	—
CIC stratified[b]	VX Small	0.9999	0.0003	0.8826
	XGBoost	1.0000	0.0002	**0.9416**
	Kitsune	0.6227	0.6910	—
CIC by-day†	VX Small	**0.7112**	**0.5731**[a]	0.4518
	XGBoost	0.6372	1.0000[a]	**0.8336**
	RF	0.6308	1.0000	0.4207
	MLP	0.5959	1.0000	0.4579
EMBER2024	VX Small	0.9969	0.0119	0.7549
	XGBoost	**0.9981**	**0.0053**	**0.8094**
	MLP	0.9970	0.0098	0.7733

Table 2: LOAO—unseen-family detection (gate) and attribution honesty (choice abstention). The control—macro-F1 on *known* families—stays at 0.38–0.56 across folds, i.e., the family that is excluded does not degrade attribution of the others. UNSW Small.

Excluded family	gate AUROC	FPR@95	abst. (%)	*n* test
analysis	0.9992	0.0002	92.0	677
backdoor	0.9989	0.0028	63.0	583
dos	0.9941	0.0202	3.0	4,089
exploitation	0.9550	0.4177	8.0	11,132
fuzzing	0.8519	0.3899	28.0	6,062
generic	0.9957	0.0128	50.0	18,871
recon	0.9523	0.1777	28.0	3,496
shellcode	0.9688	0.1305	42.0	378
worms	0.9871	0.0642	23.0	44

Calibration (E3): all methods' in-distribution ECE stays ≤ 0.065 on UNSW-NB15 but drift to 0.39–0.45 on temporal CIC (Table 1). The policy-level implication is precise: *'Noul/gate feedbacks are calibrated'* is a claim about the training distribution; it silently degrades exactly when novel-attack pressure appears, which is the moment the claim was bought for. Section 5.5 shows the effect is *additive on top of*, not independent of, class-tilt choices.

5.2 E2 — Unknown-attack honesty

Leave-one-family-out (LOAO): train without family F, test against F's held-out rows. Table 2 reports unseen-family *detection* (gate AUROC / FPR@95 without any retraining), *attribution honesty* (how much the choice head attributes to `unknown`), and a control (macro-F1 on *known* families, which must not collapse).

UNSW: the gate generalizes. Nine of nine never-seen families are detected with gate AUROC ≥ 0.85, most above 0.95, while macro-F1 on known families is preserved (0.38–0.56). The abstention rate on the excluded family ramps 3–92%—feature-dependent, not tune-dependent, which is exactly the honesty modulation we want (an unseen-but-gatelike family still realizes `unknown` 28–92% of the time instead of being absorbed into `normal`).

CIC: the failures are structural, not accidental. On stratified CIC-Small LOAO (Table 3), the gate remains strong where families are textually distinctive (`bruteforce` 0.967, `webattack` 0.979, `exploitation` 0.970, `dos_ddos` 0.852), but drops below chance on `recon` (0.426) and near chance on `infiltration` (0.533, $n = 5$): in 77 flow-statistics features, port scans from a datacenter look exactly like a benign port-touching client.

Scope: what a single-flow triage *cannot* see. This blindness is structural, and stating its boundary is part of the contribution: VX-One classifies *isolated* events (one flow's statistics, one binary's features). Reconnaissance is by definition a *temporal-aggregation* phenomenon—low-and-slow scans are invisible to per-event statistics and become detectable only in detection-by-behavior over time windows (event provenance, targets-per-time histograms, scanning divisibility). The perimeter architecture therefore does not rely on the per-event head for recon: Phase 1 (Suricata portscan signatures, which partially cover it) and any windowed detector sit in front of, or beside, the per-event gate (Section 5.4); when Phase 1 is payload-blind, the residual recon coverage is declared honestly rather than inflated.

The abstention threshold measured at the default is non-portable: on UNSW, $\tau_{\text{abs}} = 0.5$ abstains 3–92% on excluded families; on CIC, where calibration is saturated (Table 1, ECE ≤0.001), the same fixed threshold abstains on *zero* rows—every excluded-family event gets assigned to `normal` with high confidence. Under-saturated models are correctly selective, over-saturated ones are confidently wrong.

Sweep: a portable threshold does not exist (Figure 1). We swept $\tau_{\text{abs}} \in \{0.05..0.99\}$ over the LOAO logits already computed (no retraining). On UNSW the sweep admits operating points: τ=0.5 abstains on 37.5% of excluded-family rows with 55.0% false attribution to a known attack class, and rising to τ=0.99 yields 97.3% abstention at 0.66% false attribution—a monotone honesty/utility frontier an operator can pick from. On CIC there is *no* operating point: even at τ=0.99 abstention reaches only 17.8% of excluded-family rows; the dominant failure mode is *benign absorption* (94.5% of unseen-family events attributed to `normal` at saturated confidence), not scattering across attack classes (5.5%, the UNSW failure mode). Median max-probability on excluded rows is ≥0.999 on five of seven CIC folds (`recon`: 1.000), and the margin alternative ($p_{(1)} - p_{(2)}$, median 1.000 on `recon`) inherits the saturation: margin-based selective classification [12] fails for the same reason fixed thresholds do—post-hoc policies operate on the confidence distribution, and saturation has destroyed the information the policy needs. Rescuing a confidently-wrong model requires recalibration on in-scope data (per-window, online), not a sharper threshold; this is why abstention is a *mitigation with preconditions*, not a guarantee.

EMBER2024 drift: trained-unknown as an honesty mechanism. The EMBER run trains on weeks 0–45 and validates/tests on later weeks; family composition drifts measurably at test time.

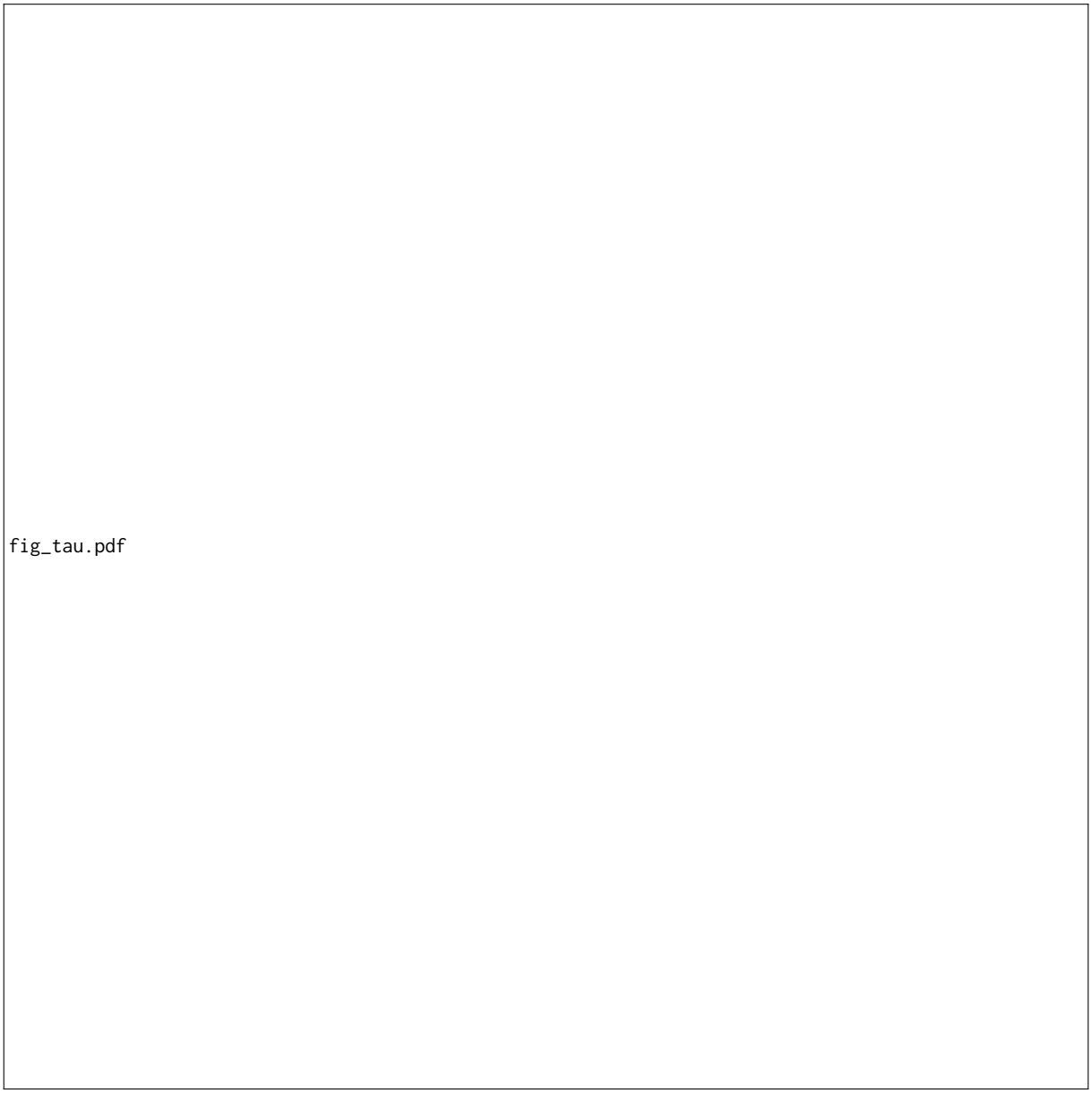


Figure 1: Abstention sweep over $\tau_{\text{abs}} \in \{0.05, 0.99\}$, mean over LOAO folds, on excluded-family test rows. UNSW: an operating frontier exists—abstention and false attack attribution cross near τ=0.5, and large τ recovers honesty (97% abstain, <1% false attribution). CIC: no operating point—unseen-family events are absorbed into `normal` (∼94%) at saturated confidence (median max-prob 1.000**; median margin** 1.000**), so no post-hoc threshold or margin policy can recover them.**

Gate-robustness is high (AUROC 0.997, FPR@95 0.012). The family-conditional table is the interesting one: *berbew*, which grew 34.5K→92K rows in test, retains recall 0.998; *wacatac*, which mutated, falls to recall 0.129 with **85%→unknown**; *grandoreiro* ($n = 226$, almost absent from train) falls to 0.221 with 78%→unknown. The model does not reduce drift into false attribution; the unknown option absorbs what the model should not claim to know, and the gate keeps the operator's safety threshold honest on the detection side simultaneously. This is the strongest single argument for the trained-unknown semantics (and contrasts directly with Table 3 on two ways of switching unknown on/off in Section 5.5).

What unknown *means under LOAO..* The two unknown mechanisms carry the same typed meaning but are not the same estimator. On EMBER2024, unknown is a trained class with real ground truth (benign and tail malware), so its probability is an in-distribution

Table 3: LOAO on stratified CIC (Small). Softmax confidence saturates: the fixed threshold abstains on nothing, and unseen families get attributed to `normal` with high confidence.

Excluded family	gate AUROC	FPR@95	abst. (%)	*n* test
botnet	0.7499	0.4038	0.00	295
bruteforce	0.9674	0.0778	0.05	2,076
dos_ddos	0.8524	0.3394	0.00	57,103
exploitation	0.9703	0.0297	0.00	1
infiltration	0.5329	0.8262	0.00	5
recon	0.4259	0.8685	0.00	23,840
webattack	0.9792	0.1550	0.00	327

Table 4: E4 latency (single-event, batch=1, 1 thread, CPU). Well within a <10 ms design budget with margin ≥ 5×.

Scale	p50 (μs)	p95 (μs)	p99 (μs)	max (μs)
Nano	675	715	773	853
Small	955	999	1,022	1,059
Base	1,652	1,795	2,082	2,335

rejection statement. Under LOAO on UNSW/CIC, the excluded family is not a candidate: abstention runs on max-softmax confidence over the *seen* families, which is out-of-distribution similarity detection [16]—the classic max-softmax baseline, with its known overconfidence failure modes—not a calibrated probability about the excluded family. We therefore report unseen-family *detection* (gate) and *abstention* separately and never describe the latter as a probability of the unseen event. Both mechanisms lose informative honesty modes measurably (Tables 2–3); the design lesson is that distribution shift has no single calibrated fix—the honest object is the pair (gate, abstention), not one number.

5.3 E4 — Latency

Measurements use single-events, batch=1, single-thread CPU, 3,000 timed events per model, on each model's exact released weights (Table 4).

Interpretation: VX-One's typed decisions carry a p50 inference cost of ~1 ms on Small, making VX-One a *per-event inline* decision component rather than a sampled-batch tool. The commercial System One class's headline latency claims are API-level; level-by-level comparison is not apples-to-apples, and we treat them as context, not as competition.

5.4 E5 — Two-phase cascade: how much of the line of fire is actually a gray zone

We reconstruct five days of CIC-IDS2017 into PCAPs (documented reconstruction, no payload— attacks' HTTP bodies are not recovered, so content-only signature rules are expected to underperform and this is the *lower bound* they encounter on this traffic in the deployed position), run Suricata 7.0.10 with fresh Emerging-Threats Open rules (52,674 of them, concatenated) plus 8 of our own structural rules to level the playing field for header-only signals, then measure per family which attacks Phase 1 catches and what lands in the gray zone for Phase 2 to handle. Table 5 collates per day.

Table 5: E5 cascade over five days of CIC-IDS2017. "F1 acc" is Phase 1 (Suricata) alone; "comb" is the estimated combined cascade; P/R = precision/recall of the combined cascade on that day's test rows. *Stats are estimates* with a documented independence assumption (Section 6).

Day	flows	attack	F1 acc	F2 acc	comb acc	comb P	comb R
Mon 03	529,450	0	0.9989	0.9996	0.9985	—	—
Tue 04	445,641	13,832	0.9681	0.9994	0.9985	0.958	0.996
Wed 05	691,390	251,722	0.6359	0.9986	0.9979	0.997	0.997
Thu 06	458,483	2,216	0.9942	0.9971	0.9961	0.552	0.978
Fri 07	702,713	288,785	0.5887	0.9979	0.9974	0.998	0.996

The headline number is the gray zone under payload opacity, not a universal estimate. Suricata with ET rules over flow-reconstructed traffic detects only ~0.1% of daily attacks by volume; the *gray zone* is 99.9% of each day *in this reconstruction*, and Phase 2 is what actually classifies it: the combined cascade retains recall ≥0.978 while precision stays ≥0.95 on every day except Thursday. We are explicit about what this experiment does and does not measure. It measures the Phase-2 recoverable share *when Phase 1 is payload-blind*—the operational regime operators of encrypted traffic (TLS everywhere, no payload inspection in managed switches) know well—and *not* the real-world gray zone of a deployment with full payload visibility, where Phase 1 resolves strictly more. Second, the cascaded numbers combine Phase 1 and Phase 2 events under a documented independence assumption at the shared-flow level; signature-triggered attacks also perturb flow statistics, so true overlap is nonzero and our combined recall is an estimate anchored at the payload-blind lower bound (Section 6). **The Thursday failure is the real lesson**: on a web-attack day of 456K benign and only 2.2K attacks, Phase 1 is structurally blind without payload (precision 0.059), Phase 2 absorbs all of it, and the FP rate of Phase 2 across 456K benign rows dominates the combined precision (0.552). A cascade is not free: *every* false flag from the blind-detection phase is multiplied by the innocent traffic around it; the correct operating posture there is Phase 2 as advisory (block with abstention, not auto-drop). This measurement—a live-tier engine's gray-zone share on a public benchmark under payload-only reconstruction—is, we believe, the crudest and most defensible novel number in this paper.

5.5 E6 — Ablations

Table 6 summarizes the ablation suite (full 14-run table and per-dataset effects in the released results). Four findings survive sanity checks and are worth their ink:

Text branch (a1). On UNSW the protocol/service/state lookups are worth ~5 macro-F1 points and nontrivial AUROC. UNSW's original feature set is partly categorical field values (proto, service, state), so the text branch supplies exactly the vocabulary a numeric encoder cannot infer. On EMBER2024 the ablation is an identity by construction: the 2,568 input features are already structural PE statistics (histograms, byte-entropy, imports), and no text field exists to ablate—which is also why the EMBER runs drop the lookup tables from the input projection entirely.

Table 6: Ablations (ref = corresponding Small reference run).

Run	Effect tested	AUROC	m-F1	ECE-ch
UNSW ref	—	0.9834	0.4571	0.0496
UNSW a1	no text branch	0.9716	0.4079	0.0509
UNSW a3	no calibration	0.9834	0.4571	**0.0980**
UNSW a4	no class tilt	0.9838	0.4545	0.0488
UNSW a6	no dropout	0.9839	0.4507	0.0451
CIC ref	—	0.9999	0.8826	0.0003
CIC a3	no calibration	0.9999	0.8826	0.0007
CIC a4	no class tilt	0.9999	**0.9313**	0.0002
CIC a6	no dropout	0.9999	0.7360	0.0007
EMBER ref	—	0.9969	0.7549	0.0076
EMBER a3	no calibration	0.9969	0.7549	0.0112
EMBER a4	no class tilt	0.9969	0.7506	0.0061
EMBER a5	unknown off	0.9961	**0.8674**	0.0022
EMBER a6	no dropout	0.9966	0.7821	0.0081

Calibration changes probability, never ranking (a3). Temperature scaling leaves argmax and AUROC untouched and improves ECE (e.g., 0.0496→0.0980 when removed, on the same logits). This is the expected signature of a well-behaved calibration layer, and worth saying plainly because calibration *is* our semantic differentiator.

Class tilts are not free (a4). On saturated CIC, removing the tilt ω *improves* attribution by +4.9 macro-F1—the de-tilt pushes minority classes *down* in a regime where the weighted cross-entropy had already pushed them up. On weakly-distributed EMBER the tilt matters less. There is no free class-tilt knob; its effect is deployment-regime-dependent, and our own loss design contains a knob whose default is wrong for saturated datasets. We say so and flag the design revision.

Dropout is dataset-dependent (a6). on CIC, zero dropout costs −14.7 macro-F1; on EMBER it *gains* +2.7 (the majority class dominates and early stopping regulates). No universal recipe; report per dataset.

Unknown is a measurable trade-off (a5-wide). Removing the trained-unknown option from EMBER's 11-way choice improves macro-F1 by +11.2 points (10-way is more balanced). "Honesty" is not free accuracy; making the `unknown` option available is a deliberate spend.

5.6 E7 — The commercial System One baseline we did not run

We planned a zero-shot API comparison (small stratified test samples billed directly against a vendor account) against Microsoft-Decision-1. It was not executed in this cycle; the paper's positioning therefore *does not lean* on beating a commercial model. What we show instead is that the class's primitive triad can be built, trained, calibrated, and stress-tested on public security data for the cost of a GPU rental, and that the evaluation discipline around it (audited labels, temporal splits, calibrated baselines, offline metric code) is entirely within reach of an academic lab. Running the vendor API remains the cheapest urgent follow-up (Section 7).

5.7 Discussion

Three threads bind the results together.

(1) The honest triage division of labor. Where the benchmark is saturable (stratified CIC), trees win attribution and a typed cascade adds nothing but a latency reduction; where the deployment regime is adversarial (temporal drift, attack families that did not exist in training), the gate/unknown semantics earn their keep in *detection robustness and attribution honesty* even while losing the calibration war (E3). The contribution is the measured boundary of where typed decisions help, not a uniform "we win everywhere".

(2) Honesty is a distinct axis from accuracy. Section 5.2 and the EMBER drift table are the clearest pair of the paper: a benchmark-optimal system (no `unknown` option, accurate operating threshold) maximizes macro-F1 while *falsely attributing* 85% of a mutated family; the honesty-first system gives up 11 macro-F1 points on the choice metric and absorbs the mutation into an honest abstention. These are different products under drift, and the macro-F1 column alone cannot distinguish them.

(3) Saturation is a deployment regime, not a benchmark property. The same dataset (CIDS2017) produces a saturated benchmark under stratified splitting and a hard unsaturated one under by-day splitting, in both ranking and calibration. Any paper reporting "F1 0.94" on IDS2017-stratified is measuring the model's ability to fit a monotone flow of re-used capture artifacts; papers should report the by-day numbers alongside (we do both).

6 Limitations

- **Severity scores are rubric-derived**, not human-labeled; the Score head's ground truth is a documented family→level mapping, so cross-level claims are approximations above benign/low/high/critical.
- **Benchmark data, audited** (Section 4): LOAO measures generalization *across benchmark families*, not robustness against real-world novel attacks; an honest reading of E2 is "the gate generalizes *more* than the choice head, on these families"—not "we detect zero-days".
- **One seed per run.** Differences under 1 macro-F1 point (a2, a4-UNSW) are noise until we re-run with 3 seeds; we plan to. Conclusions that rest on differences > 5 points are robust to this.
- **Phase 1 is a reference implementation** (Suricata + ET Open + 8 structural rules), not our C++ engine; the claims of Section 5.4 are about the *architecture*, and the reconstruction means payload-content rules are at a lower bound by construction. The "combined cascade" result is a documented two-sample estimate under an independence assumption that is exact for full-family-invisibility cases and optimistic where Phase 1 fails on exactly the stealthiest subset; we state it as an estimate, and the per-day F1/gray-zone table it rests on is a real measurement.
- **Calibration under drift** (ECE 0.40 on temporal CIC) is measured but not repaired; per-window online recalibration is the immediate follow-up and we expect the fix to be straightforward but non-trivial (the val window lies about the future in a measurable way).

- **Conformal wrappers are scoped, not promised.** Exact distribution-free coverage applies only under exchangeability: on the official UNSW split and stratified CIC, split-conformal coverage is exact [28, 35]; on every temporal split, family-composition drift is real and measured, and coverage degrades at a rate bounded by the calibration–test total-variation gap [4]. A domain classifier certifies that gap only *from below* ($d_{\mathrm{TV}} \geq 1 - 2\,\mathrm{BER}$; Appendix A)—under drift we report empirical weekly coverage with exact binomial intervals and weighted variants [35], and make no distribution-free coverage promise.
- **Cost-per-million** is not yet measured from real billing data; latency, not cost, is the number we claim.
- **Text-field attack surface** (indirect prompt injection through `user_agent`, URIs) is real [13]; V0.1 has no sanitization layer and our text-branch vocabulary lookup (bounded 135-slot vocabularies, not free max-k) is only partial mitigation.
- **Vendor-API baseline not run** (Section 5.6); the paper's claims do not currently compare against the commercial System One models it cites.

7 Conclusion

We built, trained, calibrated, and stress-tested a typed decision engine for perimeter triage—the first published instantiation and public-data evaluation of the *System One* primitive triad—and report the boundary of its advantage rather than a uniform win: detection robustness and attribution honesty under unknown families and drift, a measurable gray zone of a live signature engine, and four cleanly-measured negative results (calibration collapse under drift, non-portable abstention thresholds, regime-dependent class tilts, unknown-option accuracy cost). The cheapest urgent next steps are (i) the vendor-API comparison deferred from Section 5.6, (ii) per-window online recalibration under drift, (iii) 3-seed re-runs of the main tables, (iv) a payload-inclusive Phase-1 measurement to replace the lower-bound cascade estimate, and (v) risk–coverage curves for the composed cascade's own gating. The abstention sweep is no longer a next step: it is measured (Section 5.2, Figure 1) and its verdict is the paper's sharpest negative—on saturated deployments no abstention threshold recovers honesty, so selective behavior must be bought with data and recalibration, not with post-processing. All code, configs, tests, and measured results accompany the paper.

A Formal statements

Throughout, $\mathrm{cal}_A(f) = \mathbb{E}\big[\,|\,\mathbb{P}(A \mid f(X)) - f(X)\,|\,\big]$ is the ℓ_1 calibration error of an estimator $f \in [0,1]$ for event A; the binned ECE (15 bins) is its empirical estimator [19], with its known estimator caveats, which is why we co-report MCE and reliability diagrams and frame multiclass notions through the vector/classwise/confidence hierarchy [38].

Lemma 1 (closure under single-head post-processing). If head f is calibrated for an event A of its own output and $h : [0,1] \to [0,1]$ is measurable, then $\mathrm{cal}_A(h \circ f) \leq \mathrm{cal}_A(f) + \mathbb{E}\,|h(f(X)) - f(X)|$. *Proof.* Let $m(v) = \mathbb{P}(A \mid f(X){=}v)$. Tower property: $\mathbb{E}[\mathbf{1}_A \mid h \circ f] = \mathbb{E}[m(f) \mid h \circ f]$. Jensen: $\big|\mathbb{E}[m(f) \mid h \circ f] - h(f)\big| \leq \mathbb{E}\big[\,|m(f) - h(f)|\,\big|\,h \circ f\big]$. Triangle: $|m(f) - h(f)| \leq |m(f) - f| + |f - h(f)|$; take expectations. □

So de-tilt, temperature scaling, and thresholding do not destabilize a gate's calibration beyond the size of the transformation; the practical consequence is that each derived gate's calibration error is *bounded by its own head's measured deficiency*, and E3 measures each gate's ECE directly anyway.

Lemma 2 (CORN chain rule: exactness and degradation). With true conditionals $\pi_1(x) = \mathbb{P}(\mathrm{sev} \geq 1 \mid x)$ and $\pi_2(x) = \mathbb{P}(\mathrm{sev} \geq 2 \mid \mathrm{sev} \geq 1, x)$: (i) $\mathbb{P}(\mathrm{sev} \geq 2 \mid x) = \pi_1(x)\,\pi_2(x)$ *exactly*—CORN conditions the second step on sev ≥ 1, so there is no hidden independence assumption; (ii) for estimates (q_1, q_2), $\mathrm{cal}_{\mathrm{sev}\geq 2}(q_1 q_2) \leq \mathbb{E}\,|\pi_1 - q_1| + \mathbb{E}\,|\pi_2 - q_2|$. *Proof of (ii).* $\pi_1\pi_2 - q_1 q_2 = \pi_2(\pi_1 - q_1) + q_1(\pi_2 - q_2)$; triangle inequality with $0 \leq \pi_2, q_1 \leq 1$. The bound is on *conditional deviations*, not marginal ECEs (conditioning on extra information changes the base rate [18]); hence the drop gate's ECE is measured directly rather than inferred from its conditional heads. □

Proposition 3 (no-go: marginal ECEs do not compose). Let X be trivial and $f_1 \equiv a$, $f_2 \equiv b$ be perfectly calibrated for events A, B with $\mathbb{P}(A){=}a$, $\mathbb{P}(B){=}b$ (cal = 0 for both). The naive AND policy $g = f_1 f_2 = ab$ estimates $A \wedge B$ with $\sup \big|\mathbb{P}(A \wedge B) - ab\big| = 1/4$, attained at $a = b = \frac{1}{2}$ with $A{=}B$ or $A \cap B{=}\emptyset$ (the joint law is free within the Fréchet bounds), and identically for the OR variant. Therefore no non-trivial function of the marginal ECEs bounds composed-policy calibration error; the attainable case $A{=}B$—two heads estimating the same question, e.g., "is this an attack?"—is realistic in security, not pathological. This is why Section 3.4 restricts the controller to composing boolean decisions and forbids cross-head probability composition: every gate of v0.1 is a functional of exactly one head, verified rubric-reducible. □

Lemma 4 (BER–TV; drift certification is one-sided). For any two distributions P, Q, $d_{\mathrm{TV}}(P, Q) = 1 - 2\,\mathrm{BER}^*$, where $\mathrm{BER}^* = \inf_h \frac{1}{2}\big(\mathbb{P}_P(h{=}1) + \mathbb{P}_Q(h{=}0)\big)$ is the optimal balanced error (the likelihood-ratio test attains error $(1 - d_{\mathrm{TV}})/2$). Consequently a domain classifier with balanced error BER *certifies* $d_{\mathrm{TV}} \geq 1 - 2\,\mathrm{BER}$: detecting drift quantifies it and plugs into the coverage-gap bound of [4]; failure to separate certifies nothing—absence of evidence of drift is not evidence of exchangeability. □

Bayes rule with rejection (abstention thresholds). With uniform misclassification cost and escalation cost λ_r, the expected cost of an action is $C(a \mid x) = 1 - p_a(x)$ under typed outputs, so Bayes abstains iff $\min_a C(a \mid x) > \lambda_r$, i.e., iff $\max_k p_k(x) < 1 - \lambda_r$—Chow's 1970 rule as the 0–1-cost case [7, 8], with risk–coverage curves as the primary object [9, 12]. Asymmetric matrices of cost live in the controller: max-prob abstraction can deviate from full Bayes there (a one-line counterexample: $p = (0.55, 0.40, 0.05)$ with a 3-fold penalty on confusing the second class yields abstention under full Bayes but attribution under max-prob), which is why the deviation is measured on validation data in the controller's calibration report rather than assumed away.

References

[1] Ehsan Aghaei et al. 2022. SecureBERT: A Domain-Specific Language Model for Cybersecurity. *arXiv preprint arXiv:2204.02685* (2022). Springer DOI 10.1007/978-3-031-25538-0_3.

[2] Hyrum S. Anderson and Phil Roth. 2018. EMBER: An Open Dataset for Training Static PE Malware Machine Learning Models. *arXiv preprint arXiv:1804.04737* (2018).
[3] Anastasios N. Angelopoulos and Stephen Bates. 2021. A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification. *arXiv preprint arXiv:2107.07511* (2021).
[4] Rina Foygel Barber, Emmanuel Candès, Aaditya Ramdas, and Ryan J. Tibshirani. 2023. Conformal Prediction Beyond Exchangeability. *The Annals of Statistics* 51, 2 (2023).
[5] Marcus Botacin et al. 2025. EMBER2024: A Benchmark Dataset for Holistic Evaluation of Malware Detection. *arXiv preprint arXiv:2506.05074* (2025). DOI ACM 10.1145/3711896.3737431.
[6] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*.
[7] C. K. Chow. 1970. On Optimum Recognition Error and Reject Tradeoff. *IEEE Transactions on Information Theory* 16, 1 (1970).
[8] Corinna Cortes, Giulia DeSalvo, and Mehryar Mohri. 2016. Learning with Rejection. In *Algorithmic Learning Theory (ALT)*.
[9] Ran El-Yaniv and Yair Wiener. 2010. On the Foundations of Noise-Free Selective Classification. *Journal of Machine Learning Research* 11 (2010).
[10] Charles Elkan. 2001. The Foundations of Cost-Sensitive Learning. In *International Joint Conference on Artificial Intelligence (IJCAI)*.
[11] G. Engels et al. 2021. Troubleshooting an Intrusion Detection Dataset: The CICIDS2017 Case Study. In *IEEE SMC / conference variant to confirm.* IEEE 9474286.
[12] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*.
[13] Kai Greshake, Sahar Abdelnabi, Shailesh Mishra, Christine Endres, Thorsten Holz, and Mario Fritz. 2023. Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. In *AISec @ ACM CCS*.
[14] Jiatao Gu, James Bradbury, Caiming Xiong, Victor O. K. Li, and Richard Socher. 2018. Non-Autoregressive Neural Machine Translation. In *International Conference on Learning Representations (ICLR)*.
[15] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On Calibration of Modern Neural Networks. In *International Conference on Machine Learning (ICML)*.
[16] Dan Hendrycks and Kevin Gimpel. 2017. A Baseline for Detecting Misclassified and Out-of-Distribution Examples in Neural Networks. In *International Conference on Learning Representations (ICLR)*.
[17] Alex Kendall, Yarin Gal, and Roberto Cipolla. 2018. Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
[18] Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. 2017. Inherent Trade-Offs in the Fair Determination of Risk Scores. In *Innovations in Theoretical Computer Science (ITCS)*.
[19] Ananya Kumar, Percy Liang, and Tengyu Ma. 2019. Verified Uncertainty Calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*.
[20] Matthew V. Mahoney and Philip K. Chan. 2003. An Analysis of the 1999 DARPA/Lincoln Laboratory Evaluation Data for Network Anomaly Detection. In *Recent Advances in Intrusion Detection (RAID)*.
[21] John McHugh. 2000. Testing Intrusion Detection Systems: A Critique of the 1998 and 1999 DARPA Off-line Intrusion Detection System Evaluations as Performed by Lincoln Laboratory. *ACM Transactions on Information and System Security* 3, 4 (2000).
[22] Microsoft. 2026. Microsoft-Decision-1: Our Model for Fast Decision-Making. https://commandline.microsoft.com/microsoft-decision-1-model-foundry/. Announcement, 2026-10-09; accessed 2026-10-16.
[23] Microsoft Learn. 2026. Deploy and Use Microsoft-Decision-1 in Microsoft Foundry. https://learn.microsoft.com/en-us/azure/foundry/foundry-models/how-to/use-foundry-models-microsoft-decision. Accessed 2026-10-16.
[24] Yisroel Mirsky, Tomer Doitshman, Yuval Elovici, and Vitaly Shmatikov. 2018. Kitsune: An Ensemble of Autoencoders for Online Network Intrusion Detection. In *Network and Distributed System Security Symposium (NDSS)*.
[25] Nour Moustafa and Jill Slay. 2015. UNSW-NB15: A Comprehensive Data Set for Network Intrusion Detection Systems. In *Military Communications and Information Systems Conference (MilCIS)*.
[26] Alexandru Niculescu-Mizil and Rich Caruana. 2005. Predicting Good Probabilities with Supervised Learning. In *International Conference on Machine Learning (ICML)*.
[27] Open Information Security Foundation. 2026. Suricata: Open Source Intrusion Detection Engine. https://suricata.io.
[28] Harris Papadopoulos, Kostas Proedrou, Vladimir Vovk, and Alexander Gammerman. 2002. Inductive Confidence Machines for Regression. *European Conference on Machine Learning (ECML)* (2002).
[29] Vern Paxson. 1999. Bro: A System for Detecting Network Intruders in Real-Time. *Computer Networks* 31, 23–24 (1999).
[30] John C. Platt. 1999. Probabilistic Outputs for Support Vector Machines and Comparisons to Regularized Likelihood Methods. In *Advances in Large Margin Classifiers*. MIT Press.
[31] Martin Roesch. 1999. Snort — Lightweight Intrusion Detection for Networks. In *13th USENIX Conference on System Administration (LISA)*.
[32] Antoine Rosay, François Carlier, and Seyit Ahmet Camtepe. 2022. Errors in the CICIDS2017 Dataset and the Significant Differences in Performance of Machine Learning Models Between the Original and Corrected Versions. In *ACISP*. DOI 10.1007/978-3-031-31108-6_2.
[33] Iman Sharafaldin, Arash Habibi Lashkari, and Ali A. Ghorbani. 2018. Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization. In *International Conference on Information Systems Security and Privacy (ICISSP)*.
[34] Xintong Shi, Martin Cao, and Sebastian Raschka. 2021. CORN: Conditional Ordinal Regression for Neural Networks. In *arXiv preprint arXiv:2111.08251*. Cited as preprint; recheck venue.
[35] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel Candès, and Aaditya Ramdas. 2019. Conformal Prediction Under Covariate Shift. In *Advances in Neural Information Processing Systems (NeurIPS)*.
[36] TypeSafe AI. 2026. Jev: A System One Model for Decisions Inside Software. https://typesafe.ai. Accessed 2026-10-16.
[37] TypeSafe AI. 2026. Jev Documentation. https://docs.typesafe.ai/introduction. Accessed 2026-10-16.
[38] Jukka Vaicenavicius, Daniel Widmann, Carl Andersson, Fredrik Lindsten, Jacob Roll, and Thomas B. Schön. 2019. Evaluating Model Calibration in Classification. *arXiv preprint arXiv:1902.06977* (2019). Venue not confirmed; cited as preprint.
[39] Paul Viola and Michael Jones. 2001. Rapid Object Detection Using a Boosted Cascade of Simple Features. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
[40] VirusTotal. 2026. YARA: The Pattern Matching Swiss Knife for Malware Researchers. https://virustotal.github.io/yara/.
[41] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. 2005. *Algorithmic Learning in a Random World*. Springer.
[42] Bianca Zadrozny and Charles Elkan. 2002. Transforming Classifier Scores into Accurate Multiclass Probability Estimates. In *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*.